Supplementary Transient Heat Transfer Analysis

The original and adapted configurations of Problem 5-33 were analyzed under steady-state conditions. The results demonstrated an approximately 120.5% increase in heat flux after increasing the thermal conductivity of the nonmetallic layer and the convection coefficient at the air-exposed surface. Although this improvement indicates a greater steady-state heat transfer rate per unit area, it does not directly establish the time required for the wall to heat up or cool down. For this reason, a supplementary transient heat transfer formulation was developed to identify the governing equations, determine how time-dependent temperatures would be calculated, and explain the additional information required for a quantitative comparison.

Governing Transient Heat Conduction Equation

For one-dimensional transient conduction without internal heat generation, the energy equation is:

$$p \cdot c_p \frac{dT}{dt} = \frac{d}{dx}\left(k \frac{dT}{dx}\right)$$

Assuming constant thermal conductivity within each homogeneous layer:

$$p \cdot c_p \frac{dT}{dt} = k \frac{d^2T}{dx^2}$$

Dividing both sides by the volumetric heat capacity:

$$\frac{dT}{dt} = \frac{k}{p \cdot c_p} \cdot \frac{d^2T}{dx^2}$$

The thermal diffusivity is defined as: $\alpha = \frac{k}{p \cdot c_p}$

Therefore:

$$\frac{dT}{dt} = \alpha \cdot \frac{d^2T}{dx^2}$$

where:

* T = temperature (°C)
* t = time (s)
* k = thermal conductivity (W/m·K)
* ρ = density (kg/m³)
* cp = specific heat capacity (J/kg·K)
* α = thermal diffusivity (m²/s)

Thermal diffusivity describes how quickly temperature disturbances propagate through a material relative to its capacity to store thermal energy.

Transient Finite Difference Formulation

The same nine-node discretization used in the steady-state analysis is maintained, with a nodal spacing of:

$$\Delta x = 0.005\, m$$

The time derivative is approximated using a forward difference:

$$\frac{dT_i}{dt} \approx \frac{T_i^{n+1} - T_i^n}{\Delta t}$$

The second spatial derivative is approximated by:

$$\frac{d^2T_i}{dx^2} \approx \frac{T_{i-1}^n - 2T_i^n + T_{i+1}^n}{(\Delta x)^2}$$

Substituting into the transient conduction equation:

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha_j \frac{T_{i-1}^n - 2T_i^n + T_{i+1}^n}{(\Delta x)^2}$$

Rearranging to calculate the temperature at the next time step:

$$T_i^{n+1} - T_i^n = \frac{\alpha_j \Delta t}{(\Delta x)^2}(T_{i-1}^n - 2T_i^n + T_{i+1}^n)$$

Defining the Fourier number:

$$Fo_j = \frac{\alpha_j \Delta t}{(\Delta x)^2}$$

The explicit finite difference equation becomes:

$$T_i^{n+1} = T_i^n + Fo_j(T_{i-1}^n - 2T_i^n + T_{i+1}^n)$$

This expression applies to interior nodes located entirely within one homogeneous material. The convection boundaries and material interface require separate energy balances.

Transient Energy Balances for the Nine Nodes

For convenience, define the volumetric heat capacities of the two materials:

$$C_1 = p_1 c_{p,1}$$

$$C_2 = p_2 c_{p,2}$$

Node 0 – Steam-side boundary

$$C_1 \cdot \frac{\Delta x}{2}\frac{dT_0}{dt} = h_1\left(T_{\infty,1} - T_0\right) + \frac{k_1}{\Delta x}(T_1 - T_0)$$

Nodes 1–5 – Stainless steel interior

For i = 1, 2, 3, 4, and 5:

$$C_1 \cdot \Delta x \frac{dT_i}{dt} = \frac{k_1}{\Delta x}\ (T_{i-1} - 2T_i + T_{i+1})$$

Node 6 – Material interface

$$\left(C_1 \cdot \frac{\Delta x}{2} + C_2 \cdot \frac{\Delta x}{2}\right)\frac{dT_6}{dt} = \frac{k_1}{\Delta x}\ (T_5 - T_6) + \frac{k_2}{\Delta x}(T_7 - T_6)$$

Node 7 – Nonmetallic interior

$$C_2 \cdot \Delta x \frac{dT_7}{dt} = \frac{k_2}{\Delta x}(T_6 - 2T_7 + T_8)$$

Node 8 – Air-side boundary

$$C_2 \cdot \frac{\Delta x}{2} \cdot \ \frac{dT_8}{dt} = \frac{k_2}{\Delta x}(T_7 - T_8) + h_2\left(T_{\infty,2} - T_8\right)$$

These nine energy balances account for conduction between adjacent nodes, convection at the exposed surfaces, and thermal energy storage within the composite wall. At steady state, all time derivatives become zero, reducing the governing balances to the steady-state equations used in the original numerical analysis.

Matrix Formulation

The nine transient equations can be expressed in matrix form:

$$M\ \frac{dT}{dt} + KT = \ f$$

where M is the thermal capacity matrix, K is the thermal conductance matrix, T is the nodal temperature vector, and f contains the fluid-temperature boundary contributions.

For constant boundary conditions, the steady-state solution satisfies:

Defining the temperature deviation from steady state:

$$\theta(t)\ = \ T(t) - T_{ss}$$

Multiplying by the inverse thermal capacity matrix:

$$\frac{d\theta}{dt} = -M - 1 \cdot K\theta$$

For constant coefficients, the solution is:

$$\theta(t) = e^{-M^{-1 \cdot Kt}}\,\theta$$

This equation describes the transient temperature response of the complete nine-node system.

Mathematical Derivation of Time

To illustrate how time may be determined, consider an individual exponential thermal mode represented by:

$$\frac{d\theta}{dt} = -\lambda\,\theta$$

Separating variables:

$$\frac{d\theta}{\theta} = -\lambda\,dt$$

Integrating between the initial condition and a later time:

$$\int_{\theta_0}^{\theta(t)} \frac{d\theta}{\theta} = -\lambda \int_0^t dt$$

Solving for time:

$$\ln\left|\frac{\theta(t)}{\theta_0}\right| = -\lambda t$$

Since:

$$\theta = T - T_{ss}$$

The time expression for a single thermal mode becomes:

$$t = \frac{1}{\lambda}\ln\left|\frac{T_{inicial} - T_{ss}}{T_x - T_{ss}}\right|$$

This expression is exact for a single exponential thermal mode. It must not be treated as the general exact solution for an arbitrary node of the nine-node composite wall. The complete system has multiple thermal modes. Its characteristic decay rates are obtained from:

$$\det(K - \lambda M) = 0$$

For each node, the general solution can be written as:

$$T_i(t) - T_{i.ss} = \sum_{m=1}^{9} A_{i,m} e^{-\lambda_m t}$$

For a specified target (x) temperature, time is determined from:

$$\sum_{m=1}^{9} A_{i,m} e^{-\lambda_m t} = T_x - T_{i,ss}$$

In general, this equation must be solved numerically after the material properties, initial temperatures, and response criterion are established.

Missing Material Properties and Data Limitations

Although the original problem provides wall thicknesses, thermal conductivities, convection coefficients, and surrounding fluid temperatures, the available information is insufficient to calculate a unique transient heating or cooling time.

The following information would be required:

1. Density of the stainless steel and nonmetallic materials.
2. Specific heat capacity of both materials.
3. Initial temperature distribution throughout the wall.
4. A specific target temperature or transient response criterion.
5. The thermal boundary conditions during the heating or cooling process.

The adapted nonmetallic layer was defined through an assumed thermal conductivity of 12 W/(m·K). A specific material and its corresponding density and specific heat capacity would also need to be identified and validated before performing a numerical transient simulation. Without these data, thermal diffusivities, thermal capacity matrix values, decay constants, and numerical response times cannot be determined reliably.

Engineering Interpretation

The adapted steady-state configuration increases the calculated heat flux from approximately 8,283.18 W/m² to 18,263.41 W/m², representing an improvement of approximately 120.5%. This result demonstrates a reduction in overall thermal resistance and an increase in steady-state heat transfer capability under the specified boundary conditions. However, an increase in steady-state heat flux does not necessarily correspond to the same percentage reduction in heating or cooling time, because transient response also depends on the density and specific heat capacity of each material. The supplementary transient formulation

establishes the mathematical basis for a future heating or cooling time comparison while explicitly recognizing the limitations of the information provided in the selected textbook problem. Therefore, the improvement established by the present project is an increase in steady-state heat transfer performance, rather than a numerically verified reduction in cooling time.